

Drive Finance — Findings Guide

Plain-language summary of everything the analysis found.

Analysis date: 30 July 2026

Data: 10 CSV tables (customers, employees, loans, installments, KPIs, etc.)

Charts live in `outputs/charts/`. Full code is in `notebooks/drive_finance_analysis.ipynb`.

What the data covers

Piece	Simple meaning	Scale
Customers (users)	People who applied / borrowed	464
Employees	Sales / collection staff	104
Finance applications	Loan applications	420
Loans	Approved & disbursed loans	420
Installments	Monthly payments due	~12,300
KPI + commissions	Performance numbers	Only 10 employees

Not every employee has KPI data. When we talk about “top performers,” we only mean those 10 people.

Data quality (what to trust / ignore)

A few conventions and quirks to know when reading the numbers:

Days past due (dpd) can be negative — that is normal here

Delayed loans often show a negative `dpd` (e.g. `-29`); Active/Closed are usually 0.
This is expected coding, not a data problem. Average DPD is reported alongside loan status.

Maturity date looks wrong

It is often only ~1 month after disbursement even when tenure is 36–48 months.

Use instead: `tenure_months`.

Some empty fields are normal

- 4. CEO has no manager
- 5. Root territory (Egypt) has no parent

Unpaid installments have no payment date

ERD tables not in the CSVs

Things like system users / roles exist on the diagram but were not provided — we ignored them.

Duplicates / outliers

No duplicate rows or duplicate primary keys in any CSV.

A few installment/outstanding amounts sit outside a simple IQR fence; they were **flagged, not removed**.

Finding 1 — Overall loan portfolio health

Question: How healthy is the loan book?

Status	Share (approx.)	Meaning
Active	~2/3	Still running
Closed	~1/4	Fully paid / finished
Delayed	~10% (43 loans)	In trouble

- **Total still owed (outstanding):** about **17.7 million**
- **Average DPD:** about **-3** overall (pulled down because Delayed loans use negative DPD, e.g. -29)
- About **1 in 10** loans is Delayed — that is the main risk signal in this extract.

In one sentence: The book is mostly fine, but a clear minority is already delayed; negative DPD on Delayed loans is expected and part of how this data encodes arrears.

Finding 2 — Territories: who is big vs who is struggling

Question: Which territories over- and under-perform?

(Assumption: “over/under” means volume size and collection quality.)

Largest by money disbursed

1. **Direct Sales**
2. **Egypt**

Collection quality

- **Worst (highest overdue rate):** Direct Sales
- **Best (cleanest):** El Bahira

Takeaway

- Big volume ≠ good quality. Direct Sales is both large and stressed.
 - Smaller territories can still be quality problems even if they look small on volume.
 - Volume leaders and quality leaders are not the same list.
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Finding 3 — Who performs well, and are targets fair?

Question: Top/bottom people from KPI + commissions; are targets fair?

Coverage limit (important):

KPI and commission data exist for **only 10 employees**. We cannot rank the other 94.

What we found

- All 10 sit at roughly the **same target attainment (~87%)**.
So “who hit target” does **not** separate good from weak — targets look scaled to what people already achieved.
- **Collection rate** does separate them (best vs weakest differ by about **10 points**).
- Bigger targets tend to go with bigger books, **not** clearly with riskier books.

In one sentence: Rank these 10 by collection rate, not by “% of target”; and treat target fairness as mixed.

Finding 4 — What overdue installments look like

Question: Who is already late on payments?

- **79** installments are past their due date and not fully paid (Partial or Pending).
- Overdues concentrate in **higher-volume territories**.
- They appear across **credit-score bands** — high score does not clearly protect someone in this data.

In one sentence: Arrears are real and clustered by territory/volume, not only among “low score” customers.

Finding 5 — Early warning signals (before a loan becomes Delayed)

Question: What should we watch at the start of a loan?

Useful signals in this data:

1. **High payment-to-income (PTI)** — installment is large relative to salary (especially PTI > 0.5)
2. **High interest rate ($\geq 16\%$)** — pricing already reflects higher risk
3. **Short tenure** — shorter loans in this book co-occur with more delay

Practical rule: Flag accounts that hit **2 or more** of these at origination for early follow-up.

Credit score alone is a weak separator here (see next section).

Finding 6 — Original insight (most important)

Credit score alone is a weak predictor of Delayed loans in this dataset.

What that means in practice:

- Higher score bands do **not** cleanly show lower delay rates.
- **Affordability (PTI)** and **interest rate** carry more signal.
- A high score should not replace a payment-capacity check.

Business implication:

Refresh underwriting / early-collections priority to weight income burden and pricing, not score alone.

Simple action list

Priority	Action
1	Watch Direct Sales for collection pressure (high volume + high overdue).
2	Read negative DPD on Delayed loans as expected coding; use status, DPD, and installment arrears together.
3	Rank the KPI cohort by collection rate , not target %.
4	Build a watchlist for high PTI + high rate + short tenure (2+ flags).
5	Do not treat credit score as a sufficient risk signal on its own.

Where to look next

Want...	Open...
Short stakeholder memo	<code>outputs/executive_summary.md</code>
Charts	<code>outputs/charts/</code>
Full methods & code	<code>notebooks/drive_finance_analysis.ipynb</code>
Why these tools	<code>NOTES.md</code>
Re-run everything	<code>python -m src.run_analysis</code> from project root